

Participant Handbook

Foundations of Clinical Research · Level 1 — From Clinical Question to Publication

Provider: Neudata · #ClearDataClearImpact

Welcome

Every day you make clinical decisions that research could improve. Over **eight weeks, one session every Saturday**, this course walks the whole road of research with you: from a question at the bedside, to a sound study design, to analysing data, to writing and publishing the paper.

You do not need any research or statistics background. We go **slowly**, with pictures, examples from your own wards and a lot of group work. You will never be asked to calculate anything beyond a simple 2x2 table, and you do not need any software: the trainers run the analyses, and you learn to read and write them up.

The course at a glance

Week	Session (all on Saturday)	Theme	Your group builds
1	1	ASK — from clinical problem to research question	Proposal: topic, PICO, question, aims
2	2	FIND & READ — searching, managing references, reading a paper in 15 minutes	Proposal: search, 5 references, problem statement, working title
3	3	DESIGN — choosing the right study	Proposal: design, setting, population
4	4	BIAS & EFFECT — bias, confounding, RR/OR/NNT, diagnostic tests, forest plots	Proposal: exposure, outcomes, bias plan
5	5	DATA — CRFs, data management, describing data; <i>meet the course dataset</i>	Proposal: variables, CRF, data plan · Manuscript: Table 1 understood
6	6	ANALYSE — CIs, p-values, tests, regression, survival, missing data; <i>analyse the course dataset</i>	Proposal: sample size, analysis plan · Manuscript: results interpreted
7	7	PROTECT & WRITE — ethics, registration, integrity; writing Methods and Results	Proposal: ethics, feasibility, publication plan · Manuscript: Methods + Results

Week	Session (all on Saturday)	Theme	Your group builds
8	8	PUBLISH — Introduction, Discussion, abstract; journals, submission, peer review; presentations	Manuscript: complete draft · Proposal presented

Each session lasts **3 hours with a 15-minute break**: a short recap, two teaching blocks with frequent "check your understanding" moments, and time to build your proposal or paper. Each session also has **extra slides** at the end for anyone who wants to read further at home.

The running example

One clinical topic travels with us: **fluids in septic shock**. In Weeks 1–4 we ask, search and design the question, and look at two real papers (a randomised trial and a cohort study). In Weeks 5–8 we analyse **FLUID-ICU**, a realistic *simulated* dataset of 620 ICU patients, and write it up as a paper.

Your group and its two products

You stay in the **same group of about six** for all eight sessions. Agree roles in Session 1 (they can rotate): facilitator, scribe, timekeeper, literature lead, methods & data lead, presenter.

1. **A research proposal on your own question**, built one part every two weeks and **presented in Session 8** (6 minutes + 2 minutes of feedback). See [Assignment/proposal_brief.md](#).
2. **A draft manuscript from the course dataset**. Each group answers a different question on FLUID-ICU (e.g. fluid balance and mortality, kidney injury, ventilator-free days, lactate and missing data, ICU length of stay), using its **results pack**. You draft Methods and Results in Session 7 and the rest in Session 8. After the course, groups **review each other's drafts and reply like real authors**. See [Assignment/manuscript_brief.md](#) and [Assignment/peer_review_instructions.md](#).

Between sessions (one week; about 30–60 minutes)

After	Do this
Session 1	Think of synonyms for your PICO terms
Session 2	Finish Proposal Part 1 and the literature matrix; read the abstract of Paper 2
Session 4	Finish Proposal Part 2
Session 5	Read your group's results pack (Table 1 and figures)
Session 6	Finish Proposal Part 3; read Papers 1 and 2; start your 6-slide presentation
Session 7	Finish Proposal Part 4 and your presentation; re-read your Methods + Results
Session 8	Polish the manuscript; peer-review swap and response letter within 2 weeks

What you need

A pen, a notebook, and (optional) a laptop or phone for PubMed and the online quiz. No software to install. Participants who have done the free Neudata course *Clinical Data Analysis in R* may re-run the course analyses with the scripts in `Data/R/`.

Your toolkit

Resource	File
Research proposal template	Templates/Research_Proposal_Template.docx
Study protocol template	Templates/Study_Protocol_Template.docx
Informed consent template	Templates/Informed_Consent_Template.docx
CRF + data dictionary template	Templates/CRF_Template.docx
Literature review worksheet	Templates/Literature_Review_Worksheet.docx
Statistical analysis plan template (+ worked example <code>Data/SAP_FLUID-ICU.md</code>)	Templates/Statistical_Analysis_Plan_Template.docx
Manuscript (IMRaD) template	Templates/Manuscript_Template_IMRaD.docx
Cover letter and response-to-reviewers templates	Templates/Cover_Letter_Template.docx, Templates/Response_to_Reviewers_Template.docx
Conference abstract & poster guide	Templates/Conference_Abstract_Template.docx
Critical appraisal checklist (10 questions)	Resources/Critical_Appraisal_Checklist.pdf
Risk-of-bias tool chooser	Resources/Risk_of_Bias_Tool_Chooser.pdf
Writing & submission checklist	Resources/Writing_and_Submission_Checklist.pdf
Sample ethics application, sample grant concept note (fictional)	Resources/
2×2 and sample-size calculators	Resources/tools/
Course dataset, data dictionary, results packs	Data/
English–Vietnamese glossary, further reading	References/

The quiz

In Session 1 and again at the end of Session 8 you answer the same 24 questions. It is anonymous (participant number only) and measures what the **course** achieved, not you.

After the course

Take your proposal to the ethics committee and keep improving your manuscript. This course is **Level 1**. If you want to run analyses yourself, **Level 2** is a hands-on course in R (prerequisite: this course and the free *Clinical Data Analysis in R* course); **Level 3** offers advanced, specialised methods on request (`Course_Levels/`).

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